

# Saad Labyad

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## Education

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### University of Oxford

Oxford, United Kingdom

*PhD in Mathematics. (Grade: Awarded - Oxford PhDs are Pass/Fail only)*

Oct 2018 - Oct 2024

Thesis: "Statistical learning of Hawkes models and market microstructure".

Supervised by Prof. Samuel Cohen & Prof. Alvaro Cartea. Examined by Prof. Mathieu Rosenbaum & Prof. Mihai Cucuringu.

Funded by the Ioan and Rosemary James Scholarship and the Alan Turing Institute, affiliated with the Oxford-Man Institute.

- Developed scalable stochastic optimization methods for non-stationary Hawkes models, implemented the package `aslsd`.

- Conducted cross-sectional analysis of Nasdaq TotalView-ITCH data for high-frequency market impact modelling.

Other activities: Peer reviewer for Quantitative Finance (Taylor & Francis), co-supervisor of an MSc thesis (Hawkes Models).

Selected coursework: Algorithmic Foundations of Learning, Rough Paths and Data Science, Theories of Deep Learning.

### University of Oxford

Oxford, United Kingdom

*MSc in Mathematical and Computational Finance. (Grade: Distinction – rank top 3)*

Sep 2017 - Jul 2018

Thesis: "Topological persistence in market microstructure", supervised by Prof. Samuel Cohen.

Selected coursework: Time Series, Machine Learning, Stochastic Calculus, Algorithmic Trading, Optimization, Asset pricing.

### Ecole Centrale Paris

Paris, France

*MEng in Applied Mathematics. (Grade: 4.11/4.33 GPA – rank top 3%)*

Sep 2015 - Jul 2017

Projects: "Big data algorithms for NBO" (BEng); "Recommender systems for social networks" (MEng).

Selected coursework: Probability, Statistics, PDE, Algorithms, Computability, Risk Management, Microeconomics, Finance.

### Lycée Masséna

Nice, France

*Classes Préparatoires in Mathematics and Physics. (Grade: 176<sup>th</sup>/3370 national rank)*

Sep 2012 - Jul 2015

Undergraduate course preparing students for the national competitive entrance exams to the French Grandes Ecoles.

Selected coursework: Linear Algebra, Analysis, Multivariate Calculus, Signal Processing, Topology, Quantum Physics.

## Experience

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### Eleni

Cambridge, United Kingdom

*Quantitative Researcher (Consultant)*

Nov 2024 - Apr 2026

▪ Took time for family responsibilities, while conducting remote quantitative research work with Eleni (Cambridge Fintech).

▪ Developed and backtested cashflow forecasting pipelines on proprietary banking data, reducing RMSE by 60%.

### University of Oxford

Oxford, United Kingdom

*Teaching Assistant*

Oct 2018 - Oct 2024

▪ Taught MSc classes in Algorithmic Trading, Market Making, Stochastic Control, Stochastic Calculus, Exotic Derivatives.

▪ Conducted final mathematics interviews for applicants to the MSc in Mathematical Finance and the BA in Mathematics.

### The Alan Turing Institute

London, United Kingdom

*Doctoral Researcher*

Mar 2022 - Feb 2023

▪ Developed clustering methods for market participants in collaboration with the Authority of Financial Markets (Netherlands).

▪ Statistical analysis of proprietary Euronext limit order book data detailing the market participants behind each order.

### Ecole Centrale Paris

Paris, France

*Research Assistant*

Nov 2016 - Aug 2017

▪ Mathematical characterization of neo-classical production functions in micro-economics (Prof. Mehdi Senouci).

▪ Modelled bid-ask spreads in proprietary broker data using LSTM neural networks in TensorFlow (Prof. Ioane Muni-Toke).

## Skills

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**Methods:** Market Microstructure, Statistics, Convex Optimization, Time Series, Point Processes, Machine Learning.

**Programming:** Python (advanced – pandas, NumPy, SciPy, scikit-learn, PyTorch), C++ (intermediate), R (intermediate).

**Languages:** English (Fluent), French (Mother Tongue), Arabic (Mother Tongue), Spanish (Working Proficiency).

## Selected Publications

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▪ Labyad (2024). *Statistical learning of Hawkes models and market microstructure*. Oxford Thesis Collection.

▪ Cartea et al. (2024). *Statistical predictions of trading strategies in electronic markets*. Journal of Financial Econometrics

▪ Cartea et al. (2021). *Gradient-based estimation of linear Hawkes Processes with general kernels*. Preprint.

▪ Labyad & Senouci. (2018). *Deriving multiple-input production and utility functions from elasticities of substitution*. Preprint.

## Activities

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Label signed writer-composer, collaboration with major international music producers, sound engineering.